

Effect of diet on serum albumin and hemoglobin adducts of 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) in humans.

2-Amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) is the most abundant heterocyclic amine formed in meat and fish during cooking and can be used as a model compound for this class of chemicals possibly involved in human carcinogenesis. Knowing the exposure to heterocyclic amines is important for establishing their role in human diseases. Serum albumin (SA) and globin (Gb) adducts were first tested as biomarkers of exposure to PhIP in male Fischer 344 rats given oral doses of 0.1, 0.5, 1 and 10 mg/kg. Blood samples were collected 24 hr after treatment and PhIP released from SA and Gb after acidic hydrolysis was analyzed by gas chromatography-mass spectrometry or liquid chromatography-tandem mass spectrometry. PhIP-SA and Gb adducts increased linearly with the dose. Studies on 35 volunteers with different dietary habits exhibited that diet was a major determinant in the formation of both adducts. PhIP-SA adducts were significantly higher in meat consumers than in vegetarians (6.7 +/- 1.6 and 0.7 +/- 0.3 fmol/mg SA; respectively, mean +/- SE; $p = 0.04$, Mann-Whitney U test). The Gb adduct pattern was quantitatively lower but paralleled SA (3 +/- 0.8 in meat consumers and 0.3 +/- 0.1 in vegetarians). PhIP-SA adducts were no different in smokers and in non-smokers. The results show for the first time that PhIP-blood protein adducts are present in humans not given the synthetic compound. Both biomarkers appear to be suitable for assessing dietary exposure and internal PhIP dose and may be promising tools for studying the role of heterocyclic amines in the etiology of colon cancer and other diseases. Copyright 2000 Wiley-Liss, Inc.